Final Project: Deep Q-Networks for Pac-Man

November 10, 2025

1 Project Overview

Your goal is to design and implement the **best possible DQN architecture** for playing Pac-Man across multiple challenging layouts. Building on your homework implementation, you should experiment with advanced techniques, novel architectures, and sophisticated training strategies to create a robust agent that excels across diverse environments.

Competition Element: The team with the highest average performance will receive a single letter grade bonus. However, all teams will be graded on effort, not performance, to encourage experimentation.

2 Problem Statement

Create a DQN agent that achieves the highest possible average win rate across four distinct Pac-Man layouts:

- **classic**: Complex 19x21 maze with multiple ghosts and corridors
- **spiral**: Simple 7x7 spiral layout with predictable structure
- **spiral_harder**: Same layout but with more aggressive ghost behavior
- **empty**: Open 7x7 space requiring different navigation strategies

Challenge: Each layout presents unique strategic demands, requiring an agent that can generalize across diverse spatial structures and opponent behaviors.

3 Evaluation Methodology

Your final agent will be evaluated using a standardized protocol:

Final Score: Average win rate across all 200 games

$$\text{Score} = \frac{1}{4}\left(\text{Win rate}_{\text{classic}} + \text{Win rate}_{\text{spiral}} + \text{Win rate}_{\text{spiral\_harder}} + \text{Win rate}_{\text{empty}}\right) \tag{1}$$

Your final algorithm must work from image data alone. However, you can do whatever you want in the intermediate stages. Your code delivarables will typically include your .pt file along with your dqn_agent.py

4 Suggested Research Directions

Explore any combination of the following advanced techniques:

4.1 Network Architecture Innovations

- **Dueling DQN**: Separate value and advantage streams
- **Multi-scale CNNs**: Different receptive fields for local vs. global features
- **Attention mechanisms**: Focus on relevant parts of the game state
- **Residual connections**: Enable deeper networks
- **Custom architectures**: Design novel network structures

4.2 Advanced DQN Algorithms

- **Prioritized Experience Replay**: Sample important transitions more frequently
- **Rainbow DQN**: Combine multiple DQN improvements
- **Distributional DQN**: Model full return distribution

4.3 Training Enhancements

- **Curriculum learning**: Progressive difficulty across layouts
- **Transfer learning**: Pre-train on simpler layouts
- **Multi-task learning**: Train on all layouts simultaneously
- **Hyperparameter optimization**: Systematic search for best settings
- **Data augmentation**: Rotation/reflection of game states

4.4 State Representation

- **Frame stacking**: Multiple consecutive frames as input
- **Feature engineering**: Hand-crafted spatial features
- **Auxiliary tasks**: Predict pellet locations, ghost movements
- **Multi-modal inputs**: Combine visual and symbolic information

5 Technical Report (4-6 pages)

1. **Motivation and Related Work**: Survey of relevant DQN improvements
2. **Methodology**: Detailed description of your approach and innovations
3. **Experimental Setup**: Training procedures, hyperparameters, evaluation protocol
4. **Results and Analysis**: Performance across layouts, ablation studies, failure cases

5. **Discussion**: What worked, what didn't, and why
6. **Future Work**: Promising directions for further improvement

5.1 Presentation (20 minutes)

- Approach overview
- Key technical innovations
- Experimental results and insights
- Live demonstration of your best agent
- Q&A with class and instructor

Note: Grading emphasizes *effort*. **I am expecting 20 hrs per group member.**

6 Collaboration Policy

- All team members must contribute meaningfully to both implementation and analysis
- You may discuss general approaches with other teams, but implementations must be original
- Clearly cite any external code, papers, or resources used